



CANCER EPIGENETICS COURSE

BASIC STATISTICS

Presenter: Tran Duc Phuc

January 11th 2026

1. Descriptive statistics

Measures of central tendency

Measure of dispersion

2. Inferential statistics

Overall workflow

Parametric vs Non-parametric

Hypothesis tests

Errors

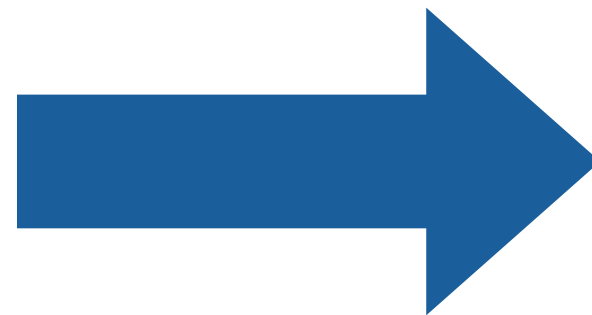
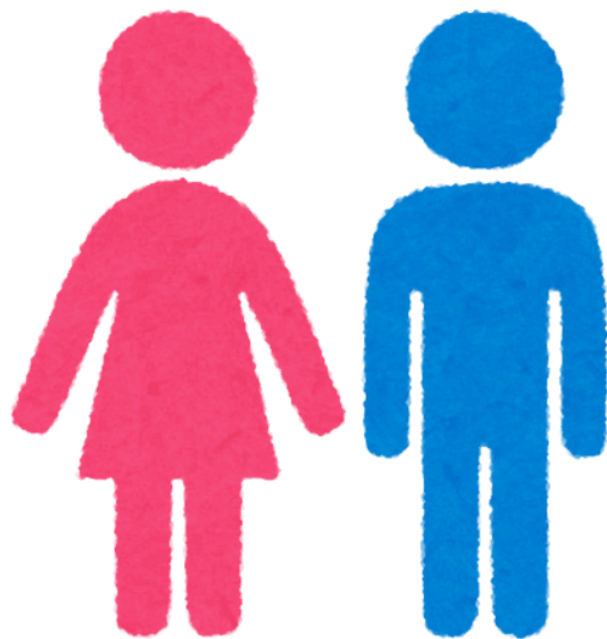
3. Correlation and Regression

1.Statistics?

Statistics: collecting, organizing, analyzing, interpreting, and presenting data to help us understand the world and make informed decisions.

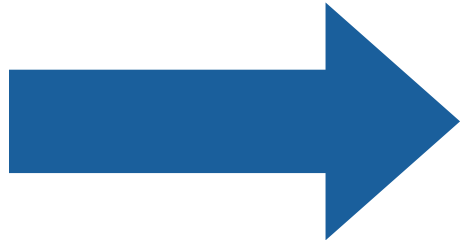
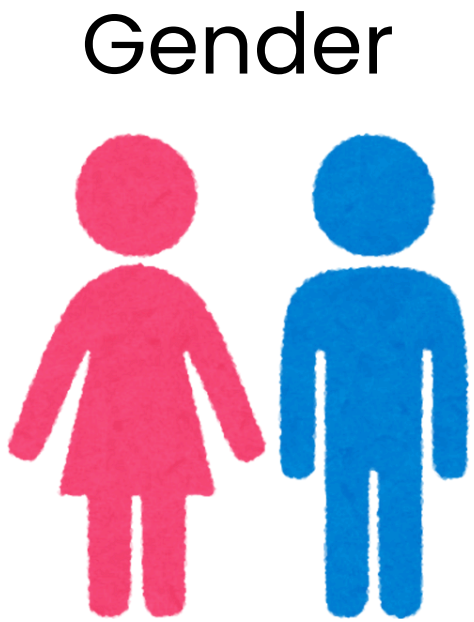
Example

Gender



Condition



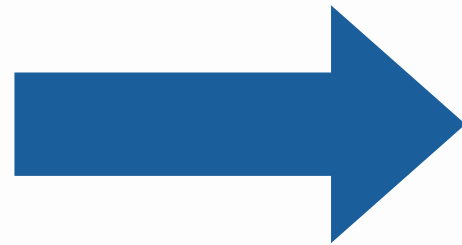


Patient ID	Gender	Condition
1	Female	Healthy
2	Female	Healthy
3	Female	Treatment 1
4	Female	Treatment 1
5	Female	Treatment 1
6	Male	Healthy
7	Male	Healthy
8	Male	Treatment 1
9	Male	Treatment 1
10	Male	Treatment 1

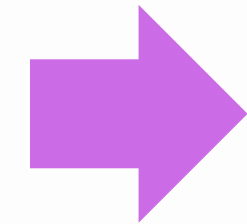
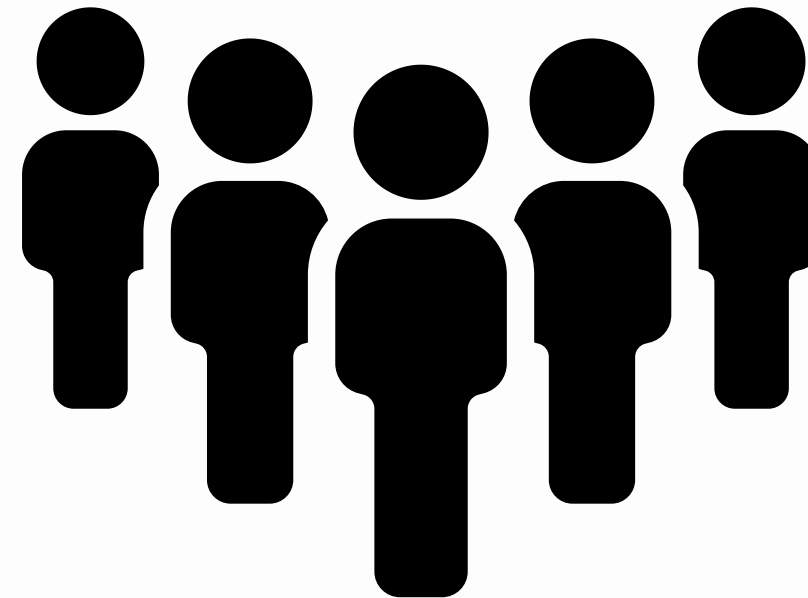
Population



Sampling

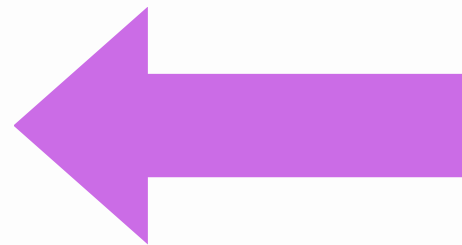


Sample

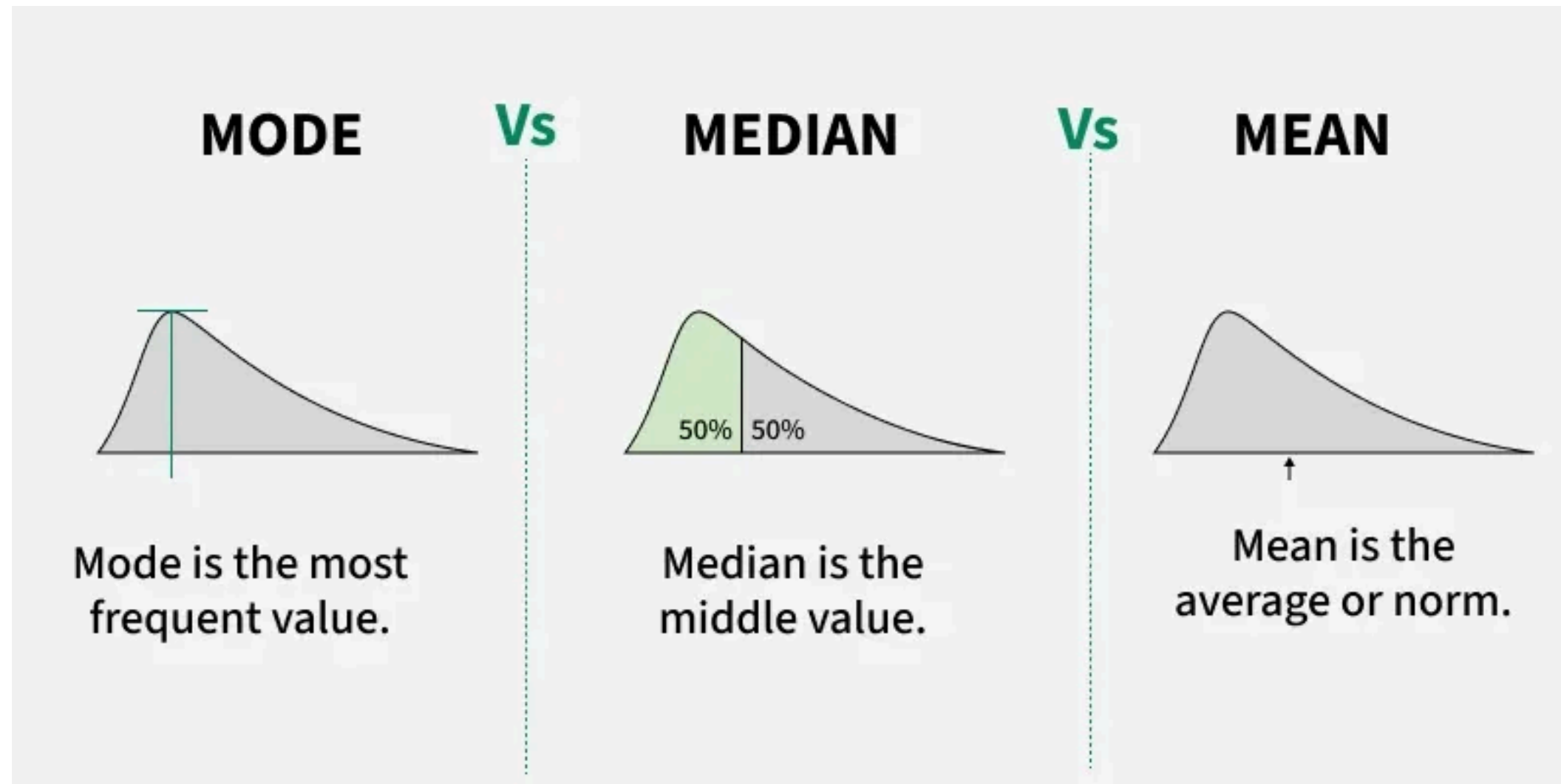


***Descriptive
statistics***

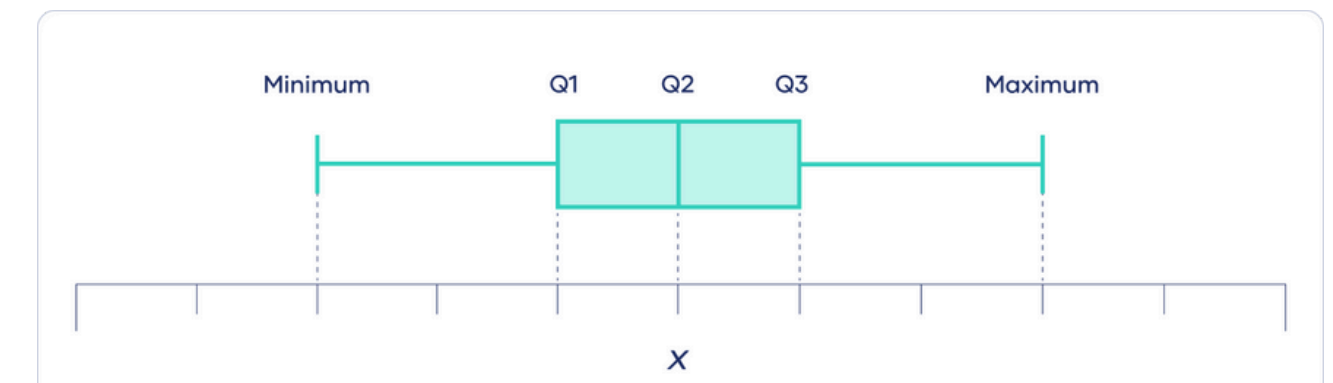
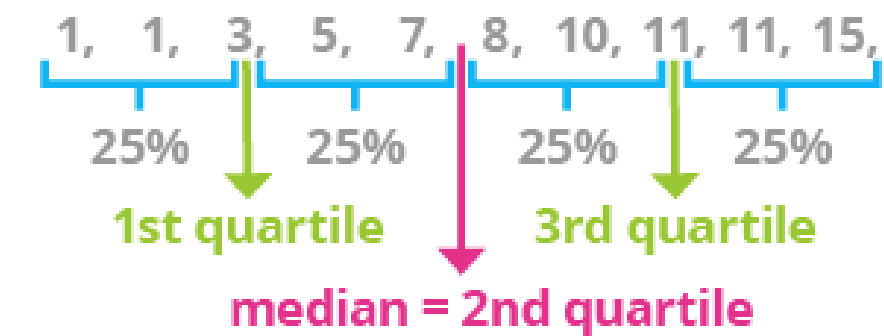
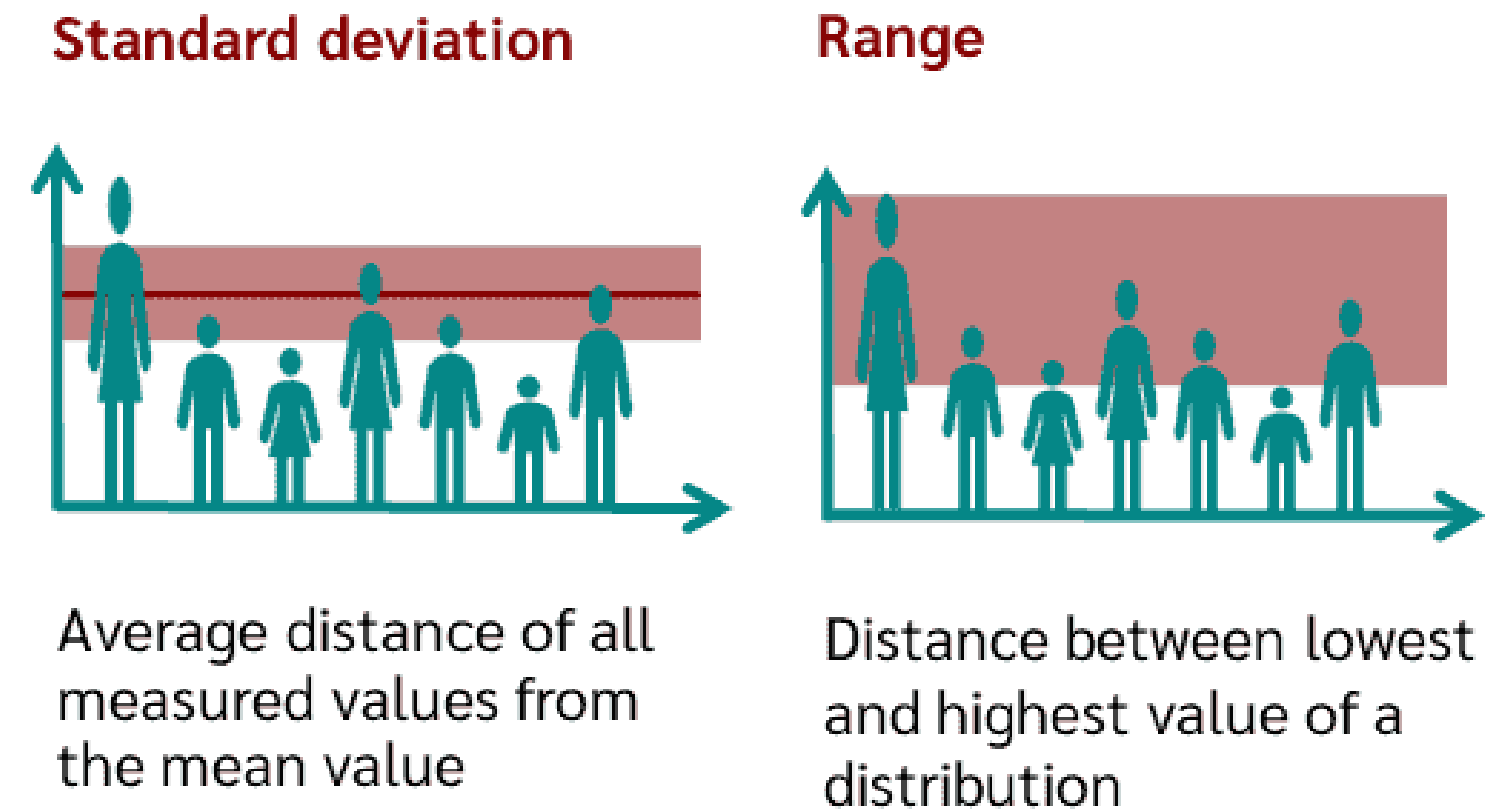
***Inferential
statistics***



Descriptive statistics

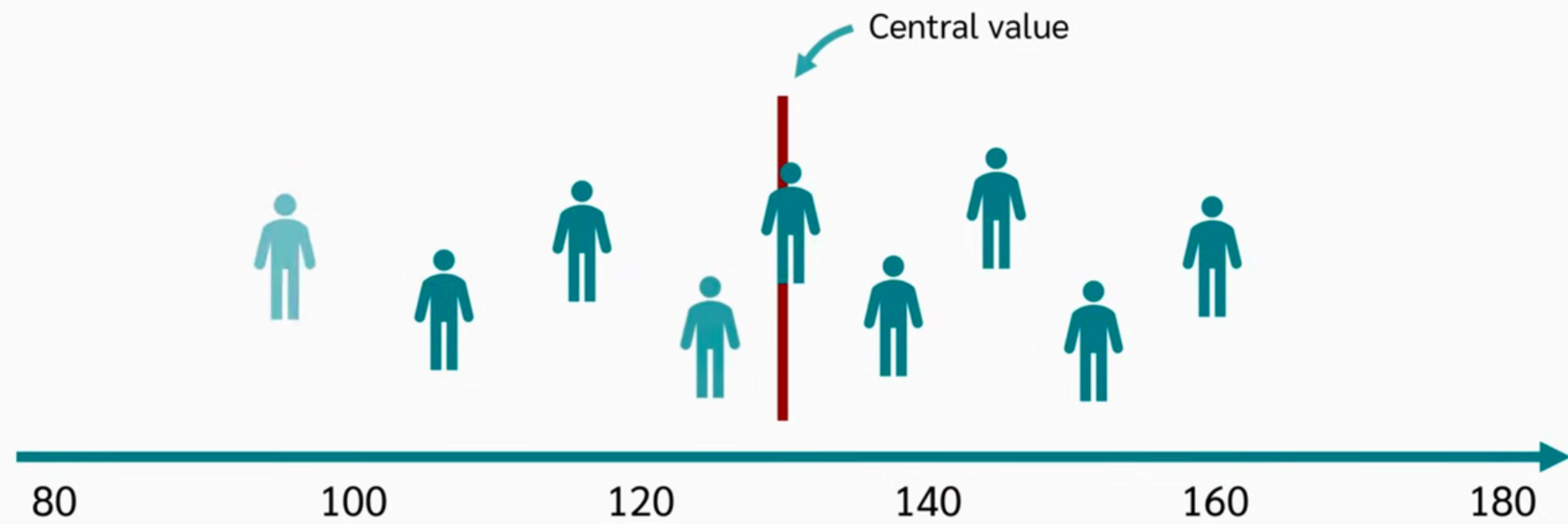


Measure of central tendency

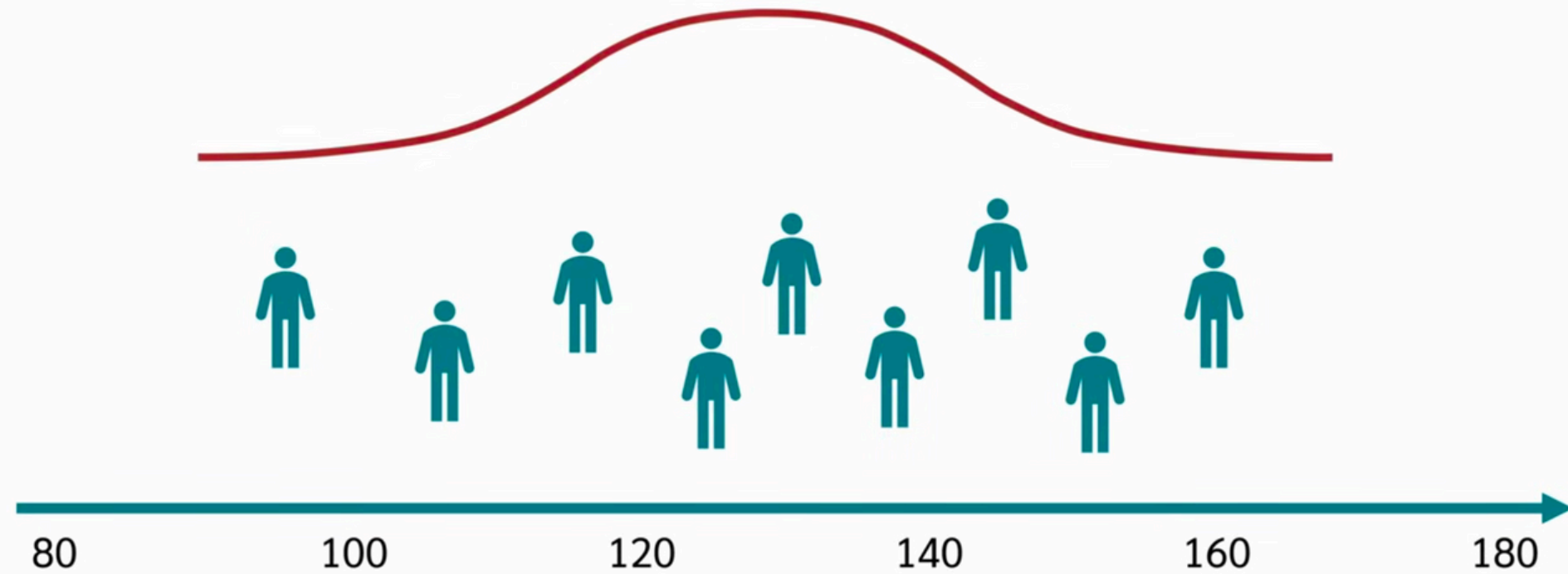


Measure of dispersion

Measures of central tendency



Measures of dispersion

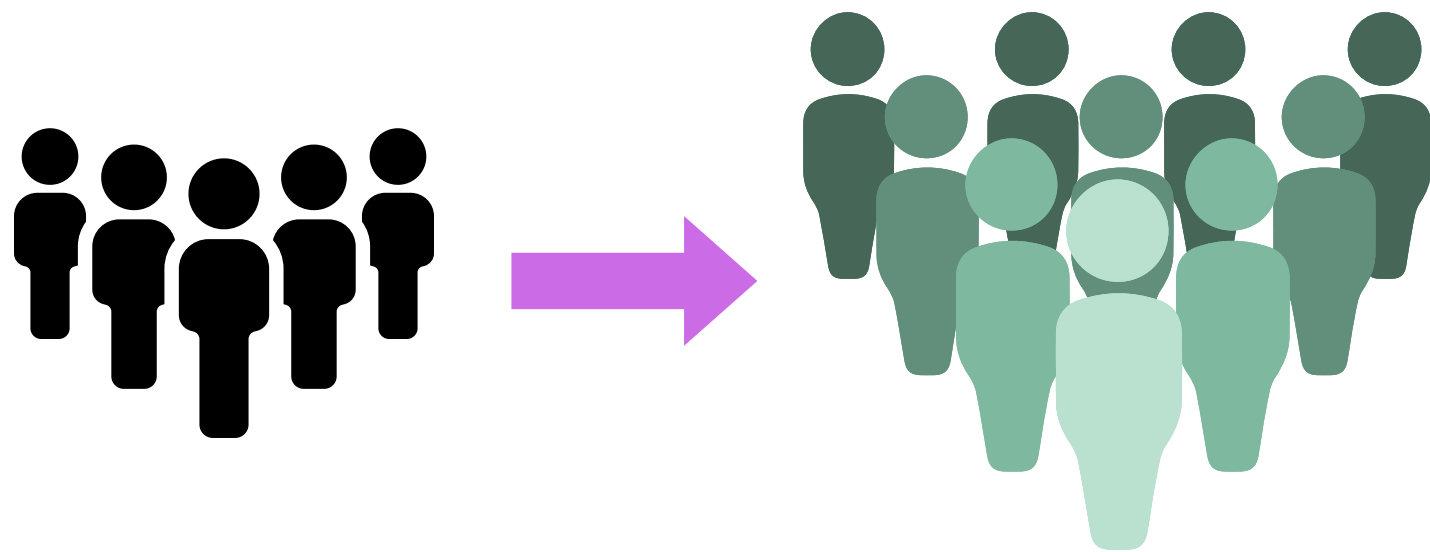


Why calculate these metrics?

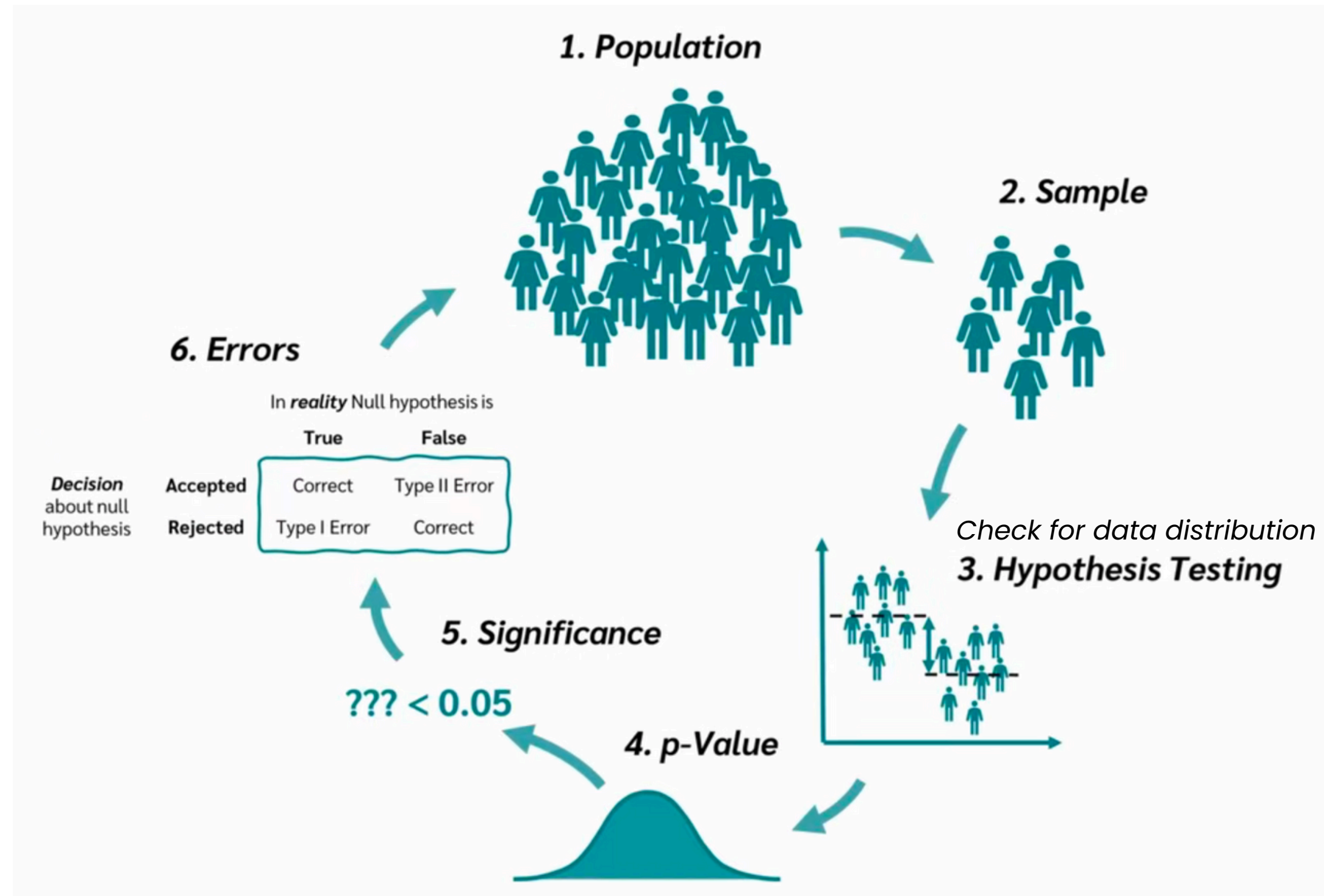
- Central tendency: identify the central point of data
- Dispersion: How the data spread

Inferential statistics

Describe, summarize
the large population based
on the collected samples

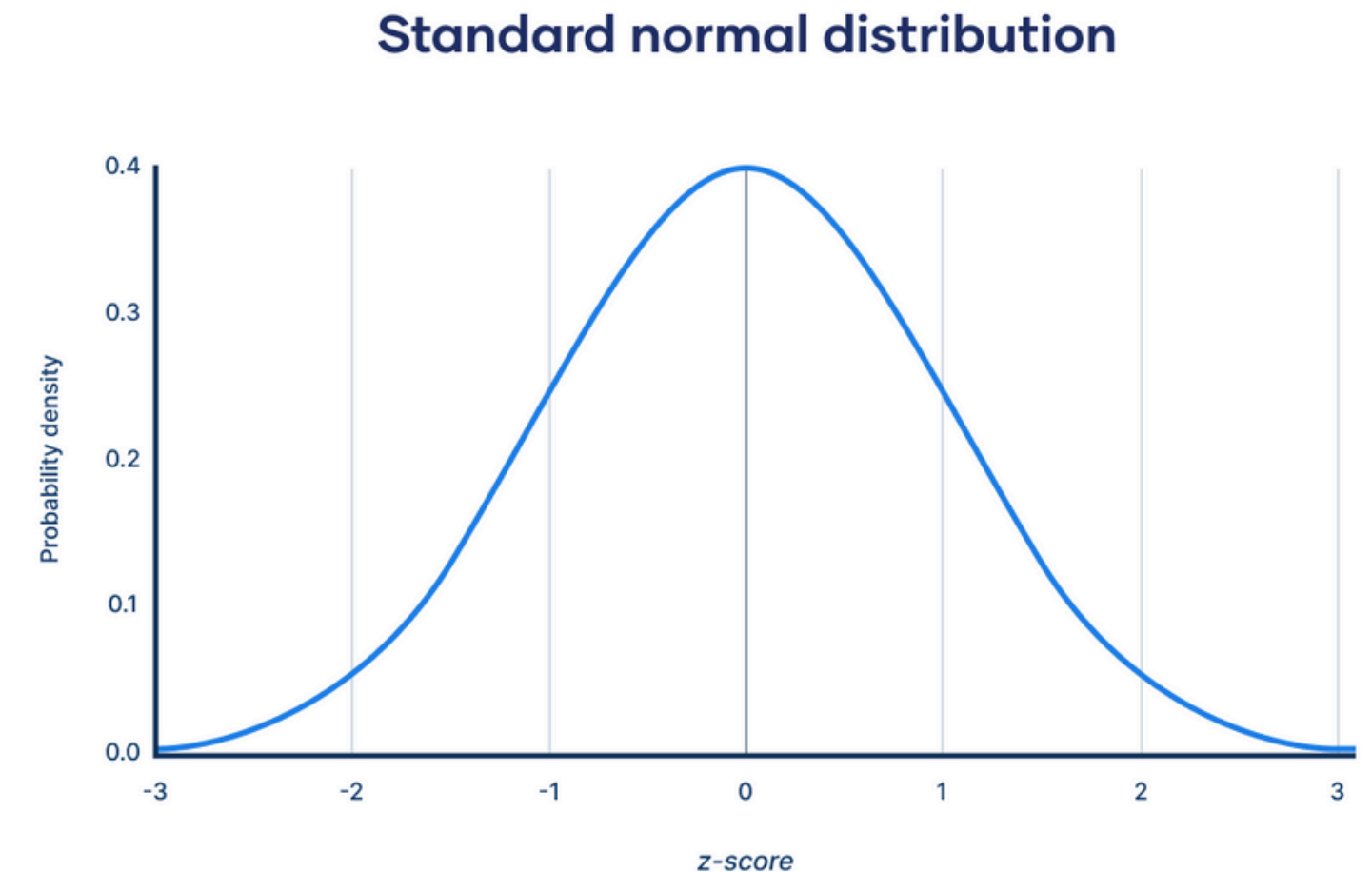


- Inferencing with multiple hypothesis statistical test
- The type, the distribution, the variable of data will determine which statistical test to use



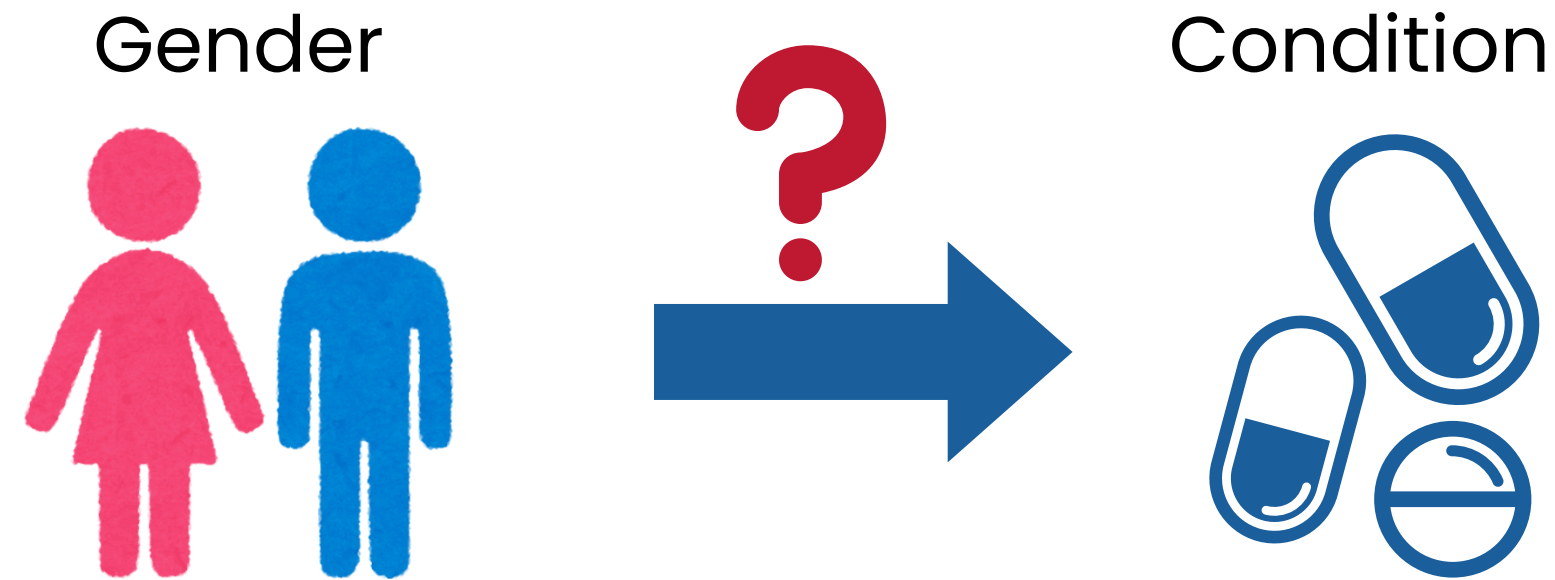
Parametric vs Non-parametric

- Data distribution: normally distributed or not?
 - Visualization method: histogram
 - Test: Shapiro-Wilk Test, Kolmogorov-Smirnov Test
- Parametric/Non-parametric test: depends on data distribution and type of variable
 - Parametric test for normally distributed data
 - Non-parametric test for non-normal distributed data

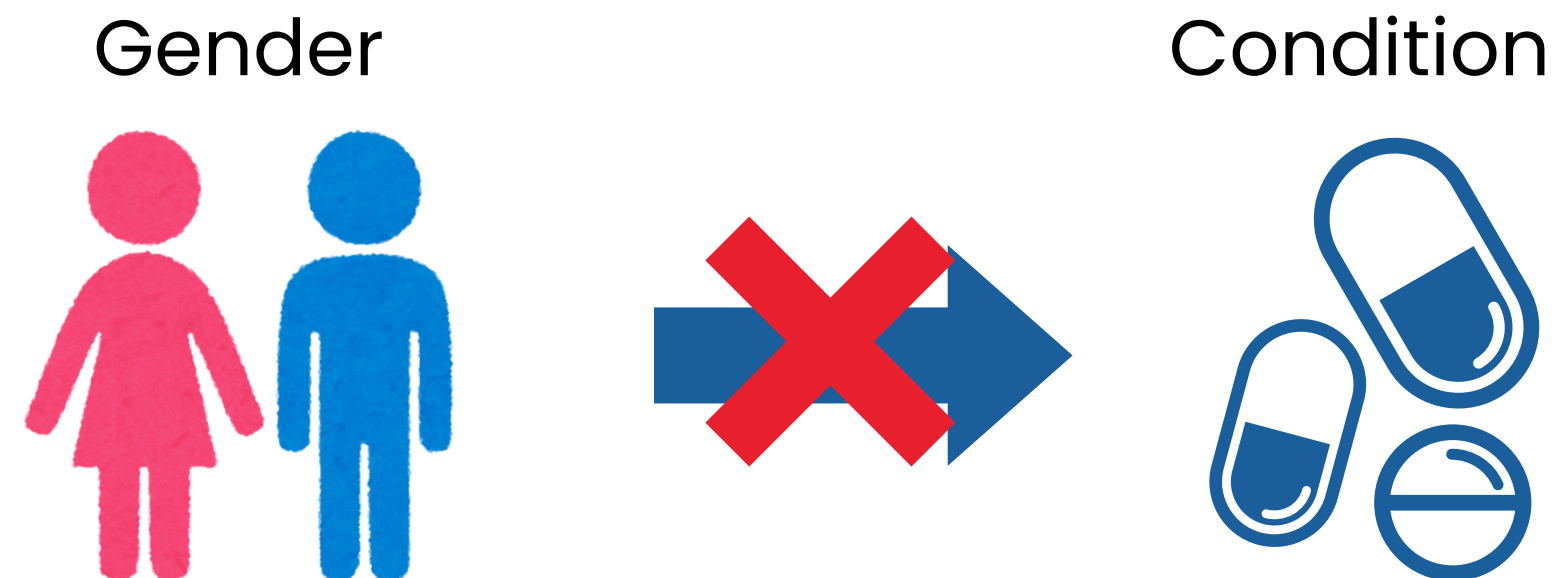


Hypothesis test

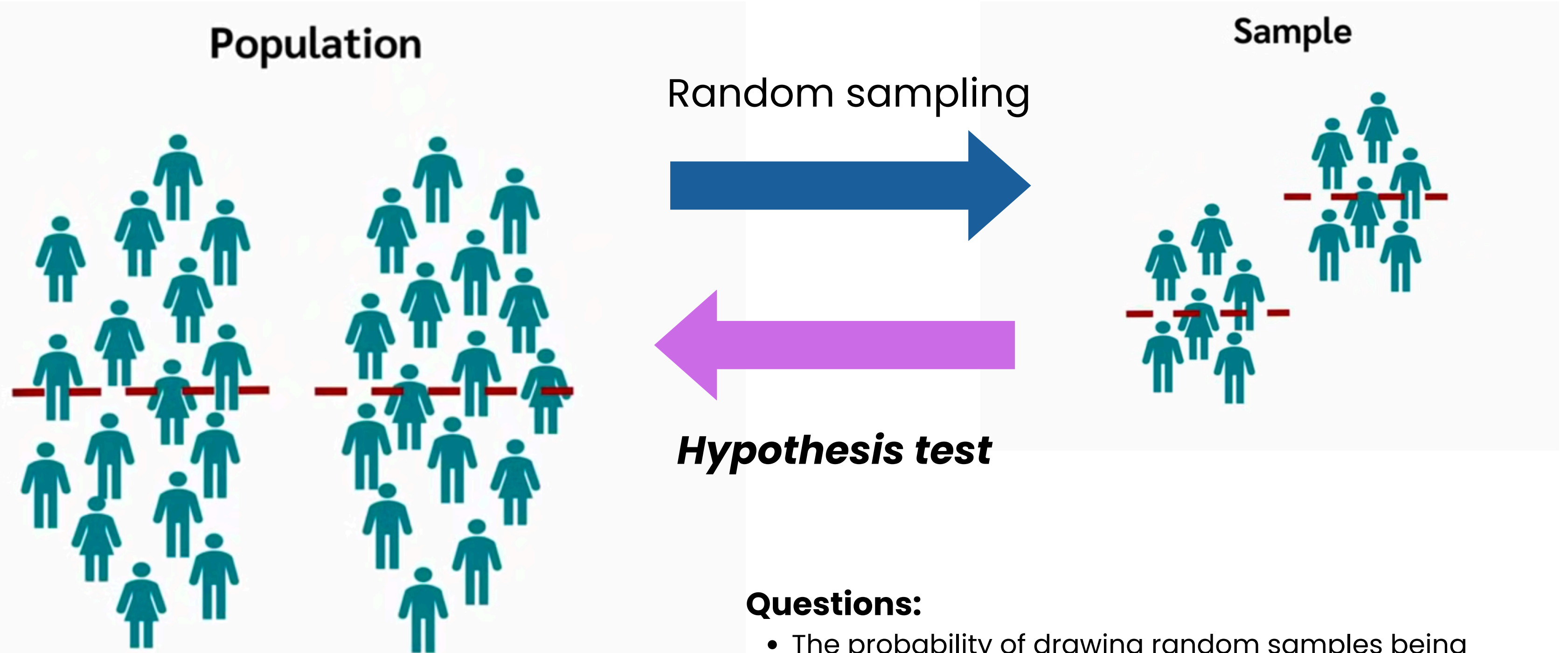
Aim: Does gender affect the condition?



Null Hypothesis (H_0): The gender has **NO** effect on the Condition in the population



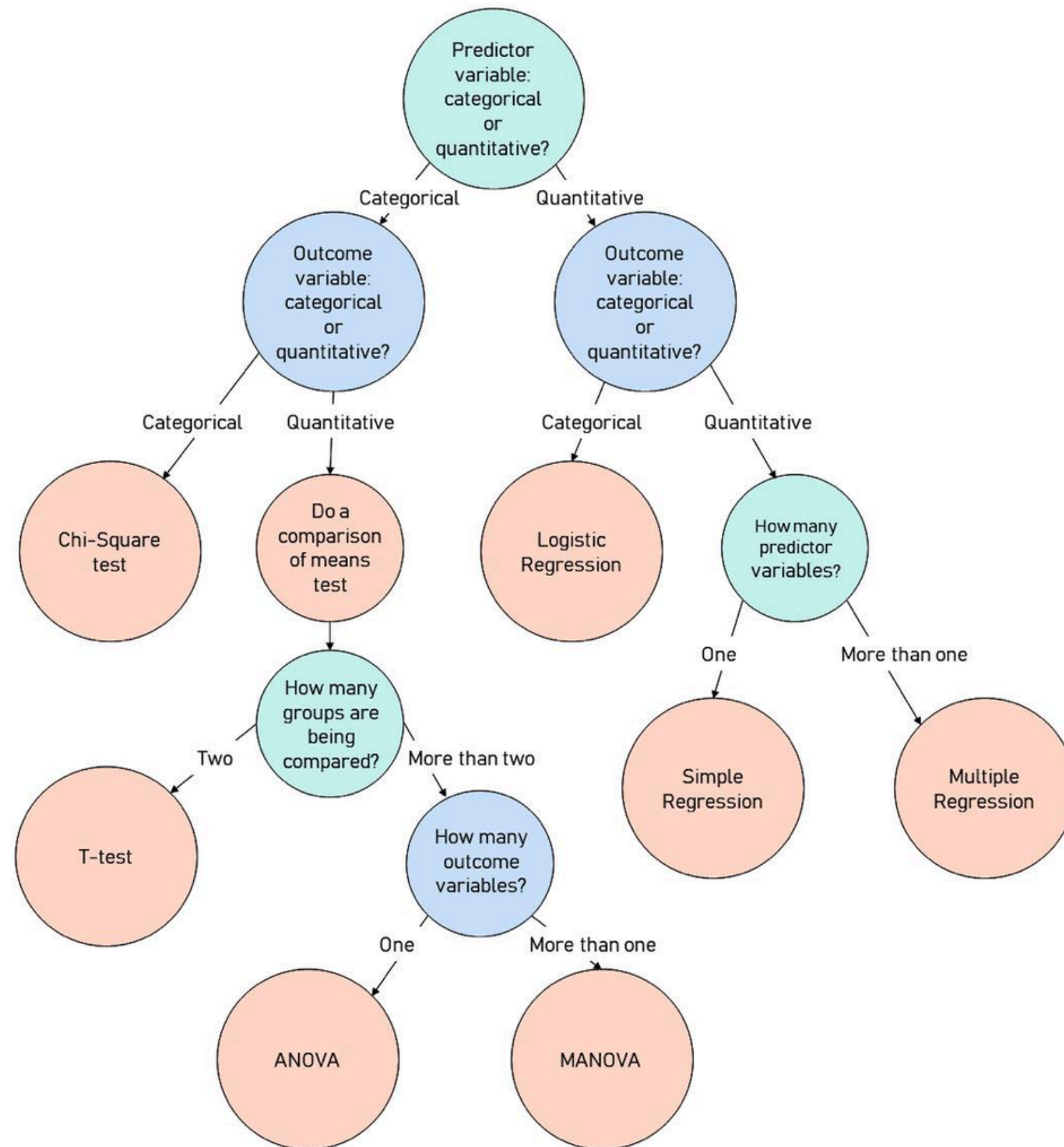
Hypothesis test



Questions:

- The probability of drawing random samples being deviated, even though the population does not have any differences
- Maybe the population has differences, enough to reject the hypothesis => ***p-value*** (< 0.05)

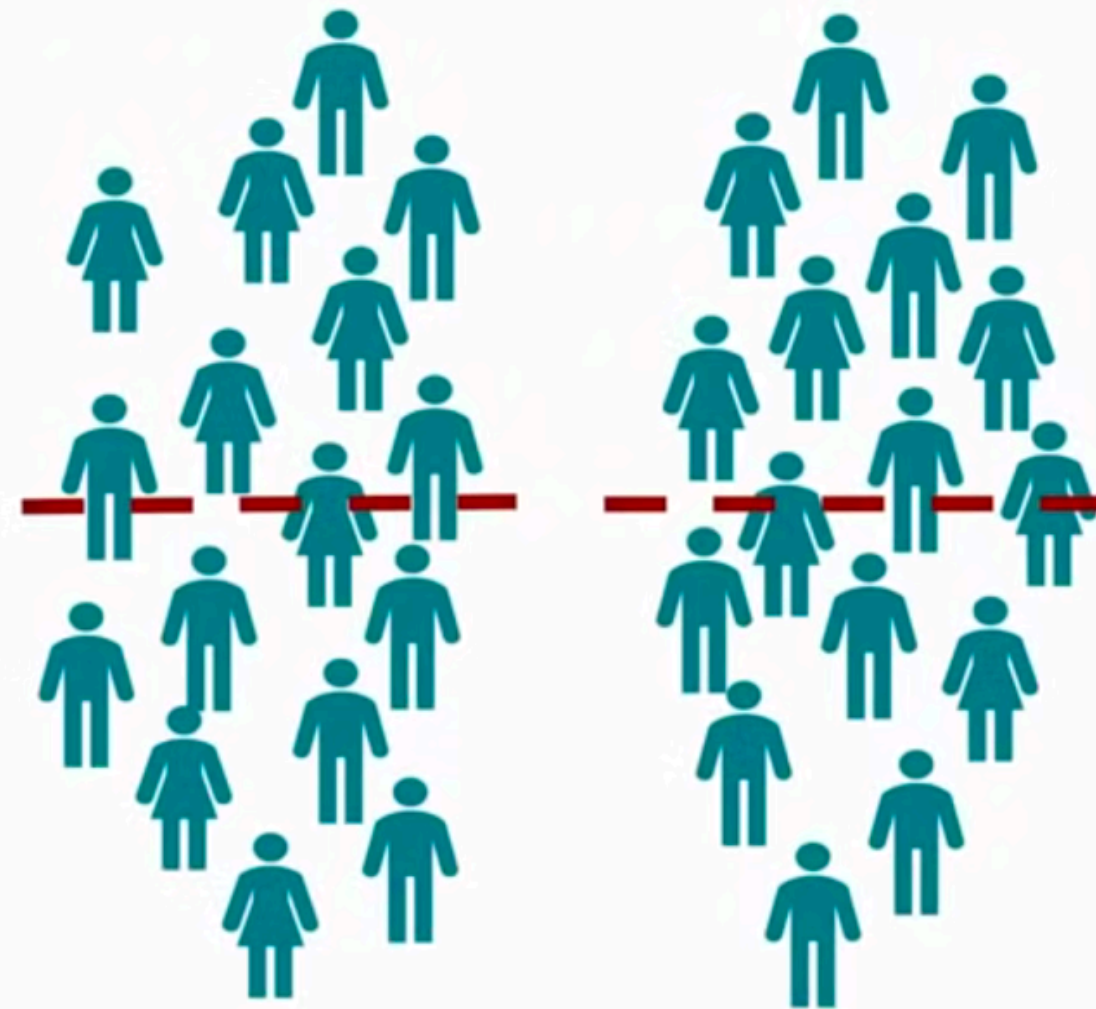
Parametric



Non-parametric

Parametric Test	Non-Parametric Version
Chi-Square	Chi-Square
T-test	Wilcoxon Rank Sum / Mann-Whitney U
ANOVA, MANOVA	Kruskal-Wallis Test
Regression	Spearman Rank Correlation

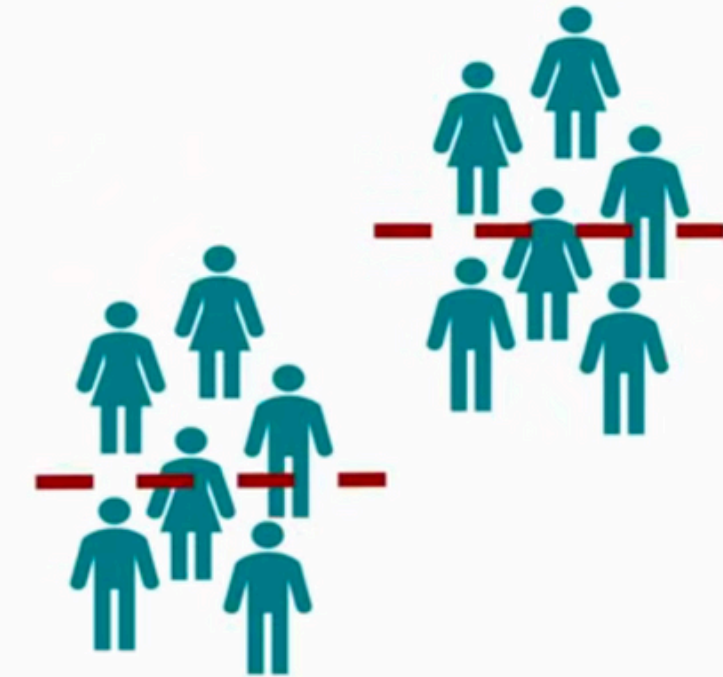
Population



Random sampling



Sample



Hypothesis test

Results of hypothesis test:

- **p-value > 0.05**: The collected sample is the same as the population (more than 95%) => the H_0 is true
- **p-value < 0.05**: The collected sample is different from the population => reject the H_0

Risk!!

- A small p-value does NOT TOTALLY prove the NULL hypothesis FALSE
- This forms 2 Errors: Type I error and Type II error

Decision	If H0 is True (No effect)	If H0 is False (Effect exists)
Reject NULL Hypothesis	Type I Error (False Positive): gender matters , but the calculation result in does not matter .	Correct Decision
Fail to Reject NULL Hypothesis	Correct Decision	Type II Error (False Negative): gender does not matter , but calculation result in matters .

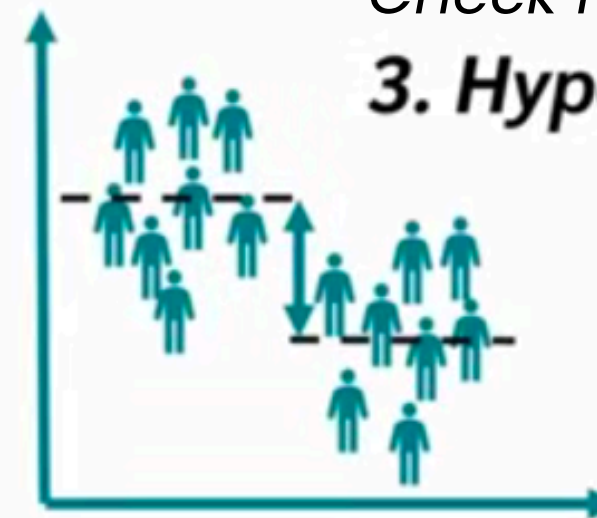
1. Population



2. Sample



3. Hypothesis Testing



4. p-Value



$$??? < 0.05$$

5. Significance

6. Errors

Decision
about null
hypothesis

Accepted
Rejected

In *reality* Null hypothesis is

True

False

Correct

Type II Error

Type I Error

Correct

Reference

